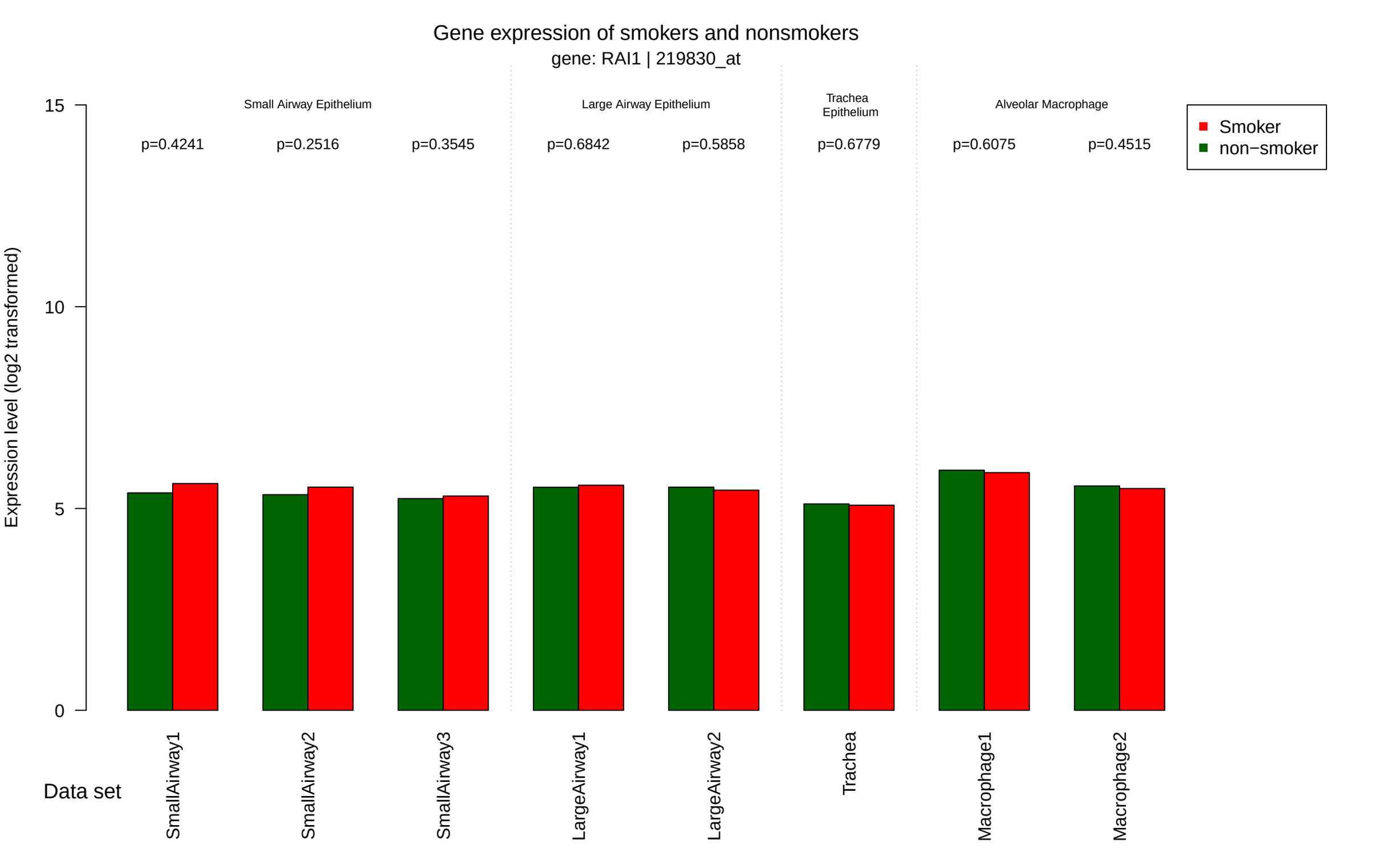
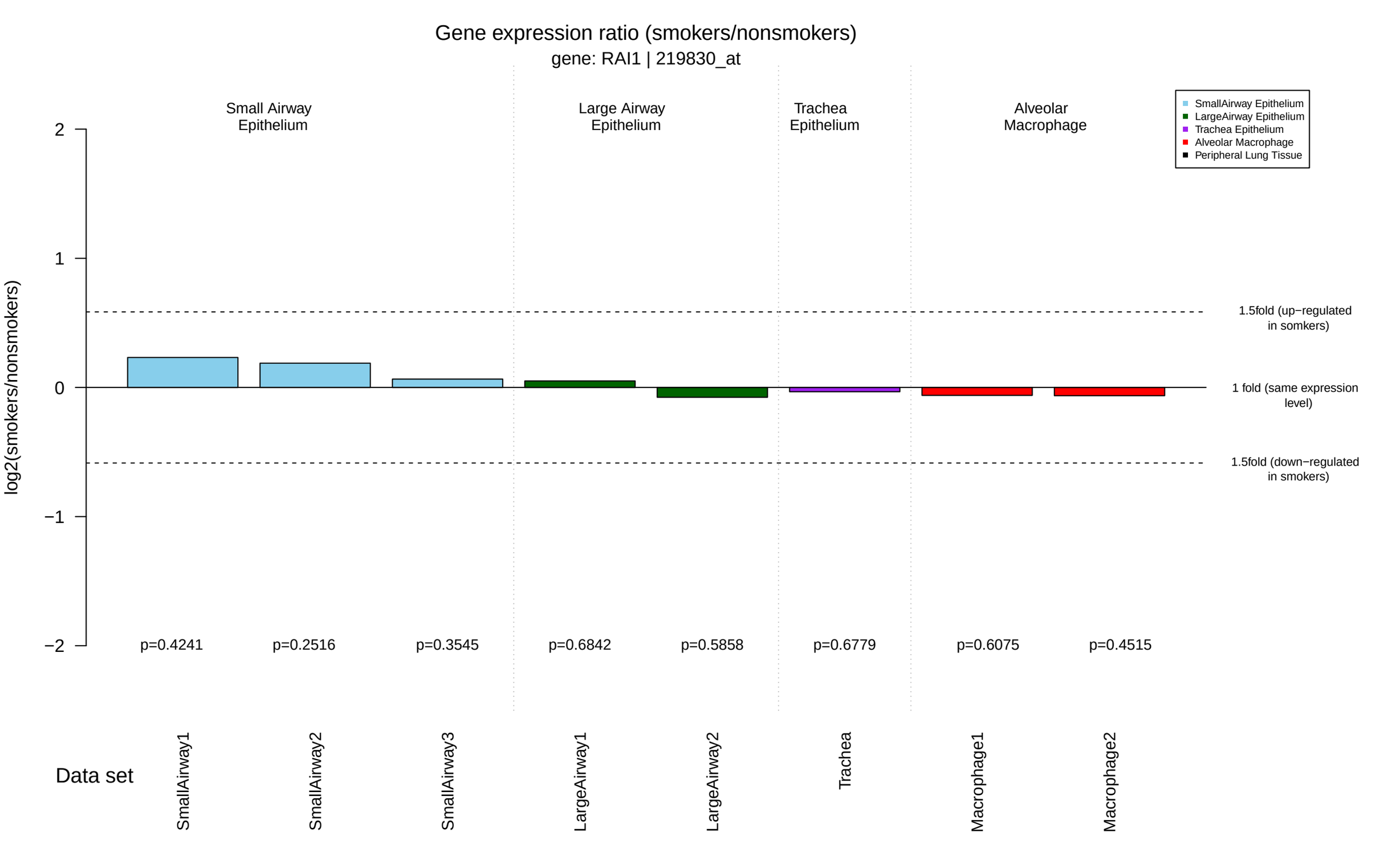
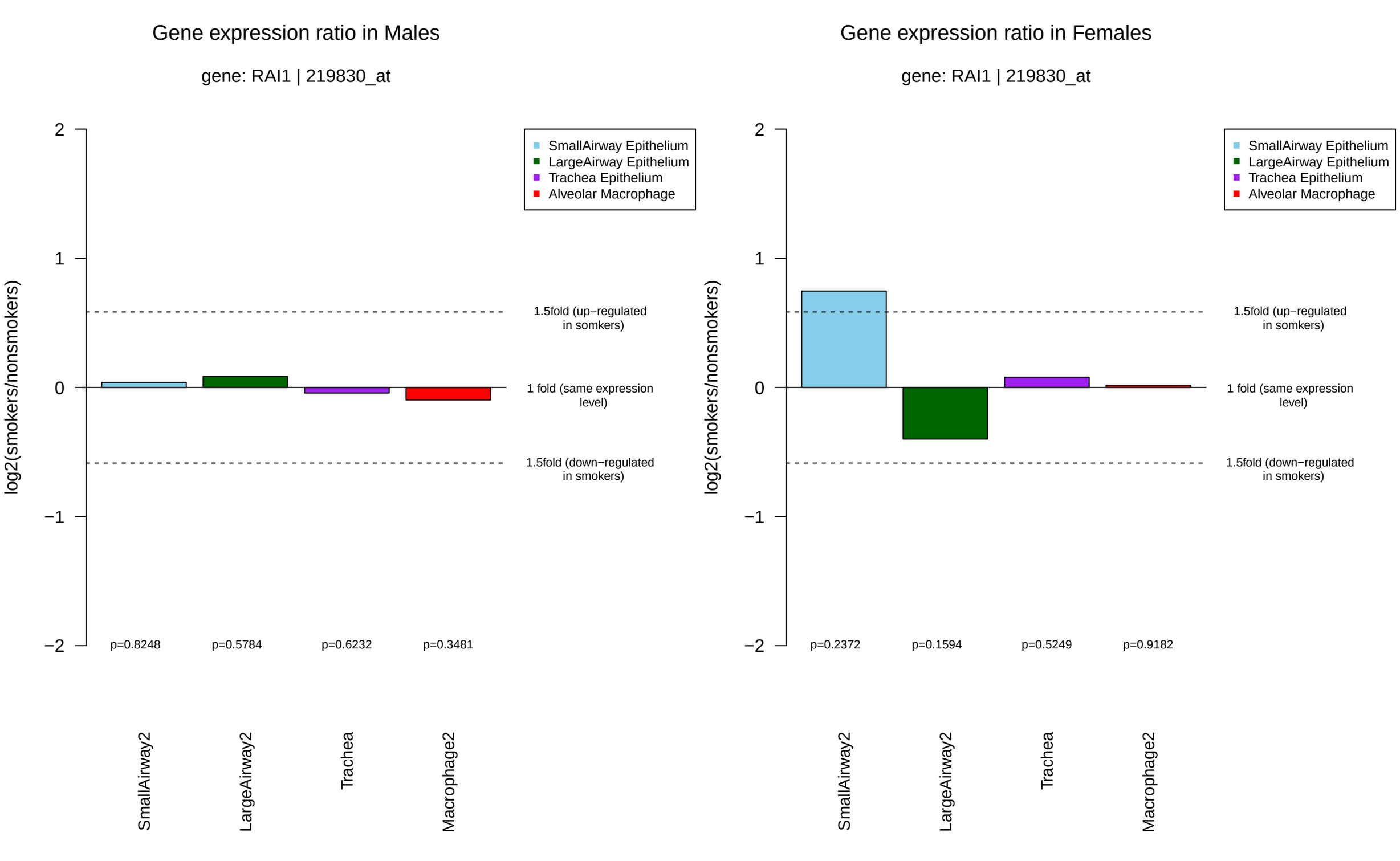
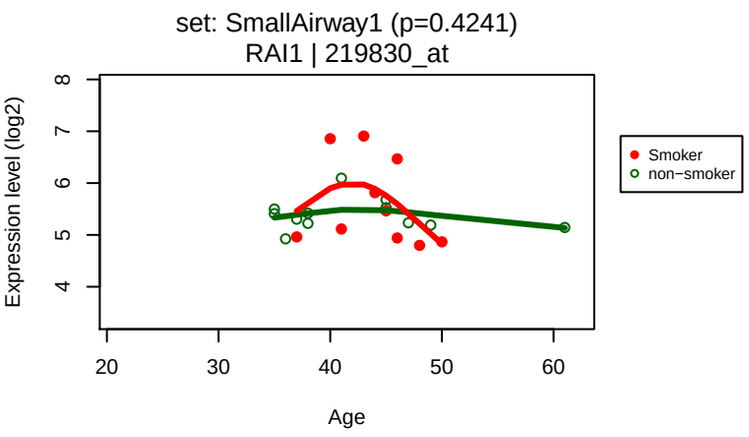
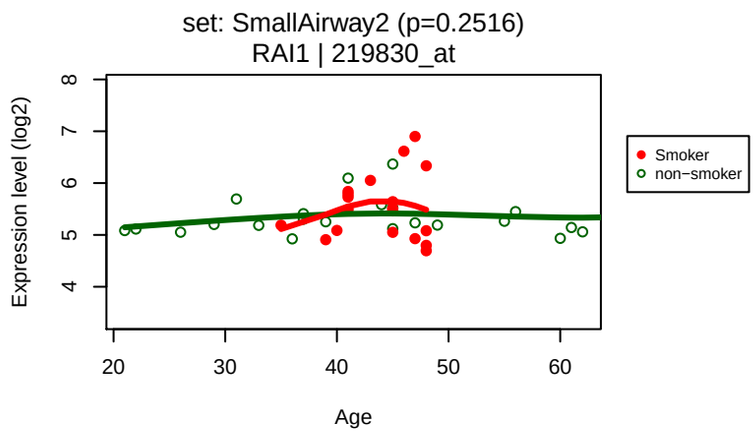
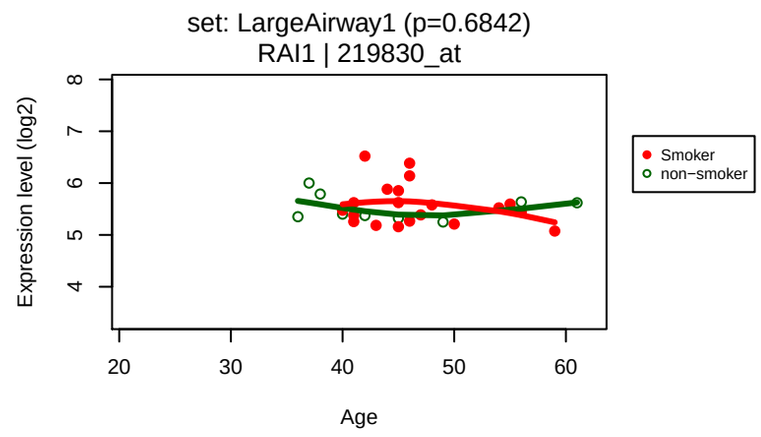

Gene expression of smokers and nonsmokers
gene: RAI1 | 219830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4241
p=0.2516
p=0.3545
p=0.6842
p=0.5858
p=0.6779
p=0.6075
p=0.4515
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

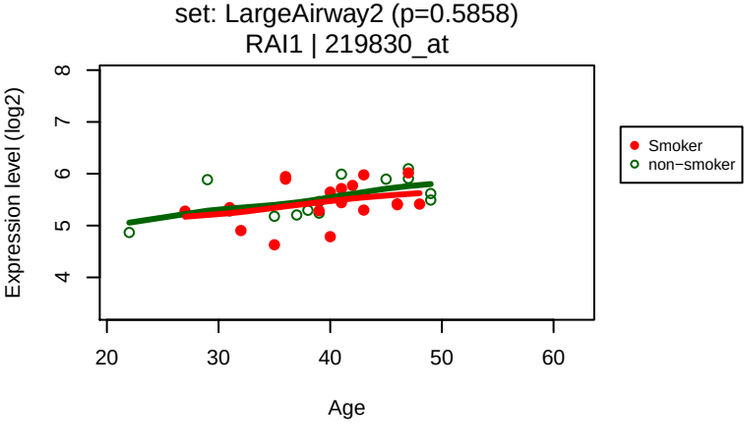

Gene expression ratio (smokers/nonsmokers)
gene: RAI1 | 219830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4241
p=0.2516
p=0.3545
p=0.6842
p=0.5858
p=0.6779
p=0.6075
p=0.4515
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

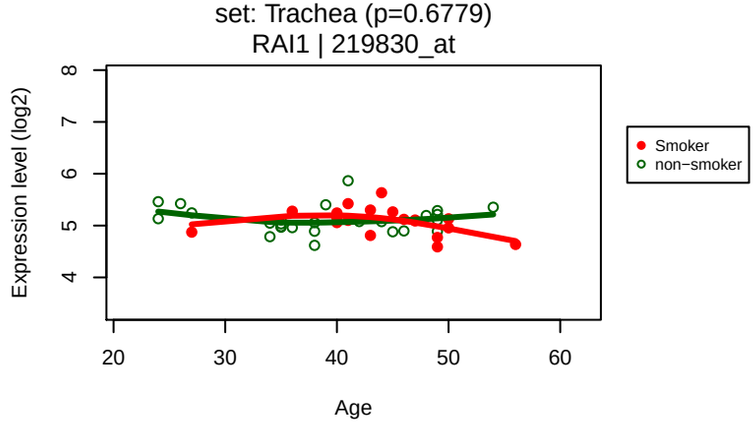

Gene expression ratio in Males
gene: RAI1 | 219830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8248
p=0.5784
p=0.6232
p=0.3481
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAI1 | 219830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2372
p=0.1594
p=0.5249
p=0.9182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

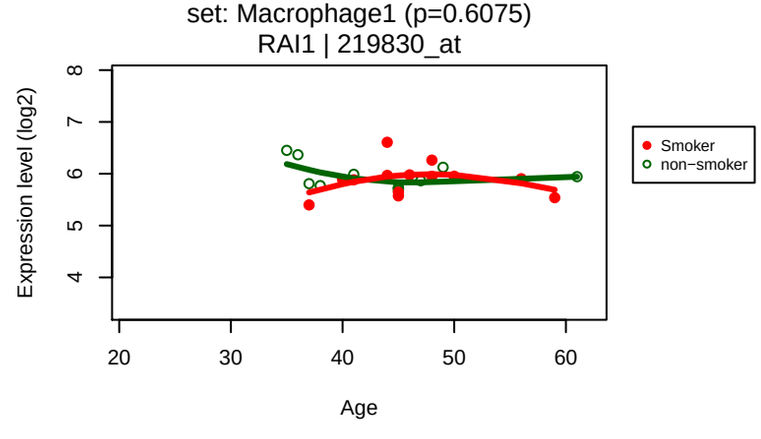

set: SmallAirway1 (p=0.4241)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker

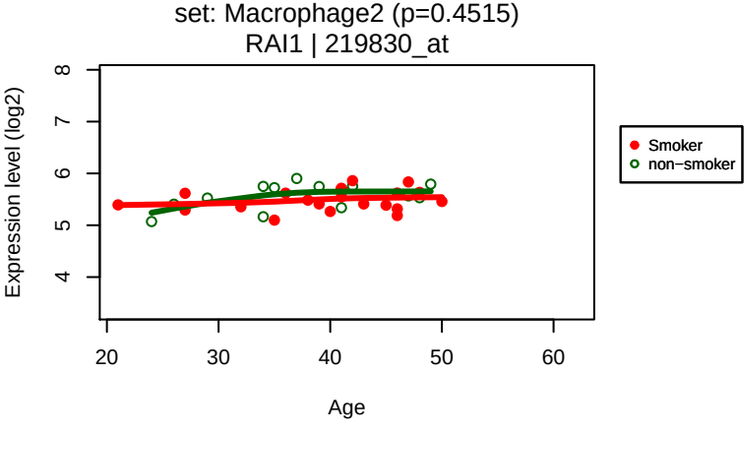

set: SmallAirway2 (p=0.2516)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker

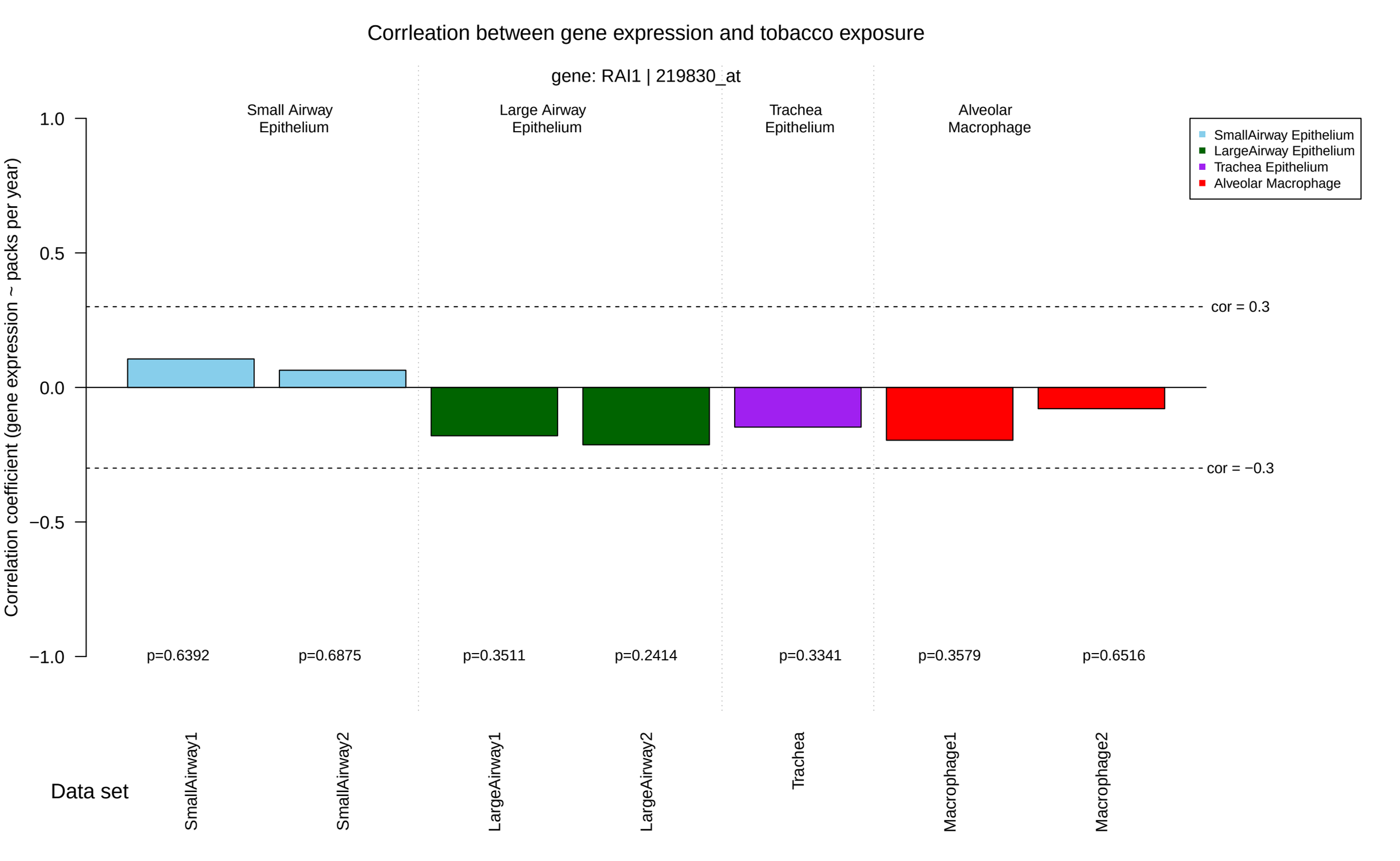

set: LargeAirway1 (p=0.6842)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5858)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6779)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6075)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4515)
RAI1 | 219830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAI1 | 219830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6392
p=0.6875
p=0.3511
p=0.2414
p=0.3341
p=0.3579
p=0.6516
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2